Last time: A m rows, n colums
rank$(A) = r$ ($=$ # pivots)

- $\dim \mathrm{col}(A) = \dim \mathrm{row}(A) = r$
- $\dim \mathrm{null}(A) = n - r$.

Lecture 16 – Determinants

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

1) $\det I = 1$

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \qquad \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

2) det changes sign if you exchange rows

$$\begin{vmatrix} c & d \\ a & b \end{vmatrix} = cb - ad = -\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

3) det depends linearly on the first row

$$\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$$

$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Remarks:
- $\det(A+B) \neq \det(A) + \det(B)$
- $\det(tA) = t^n \cdot \det A$
  (A is $n \times n$)
- All properties of the determinant are consequences of 1), 2) and 3).
  (Including the horrible formula with $n!$ terms.)

4) If A has two equal rows, then $\det A = 0$.

$$\begin{vmatrix} a & b \\ a & b \end{vmatrix} = ab - ba = 0.$$

Reason: Exchanging rows,
$\det A = -\det A$
by (2).

5) det does not change with row operations:

$$\begin{vmatrix} a - \ell c & b - \ell d \\ c & d \end{vmatrix} \overset{(1)}{=} \begin{vmatrix} a & b \\ c & d \end{vmatrix} - \begin{vmatrix} \ell c & \ell d \\ c & d \end{vmatrix}$$

$$\overset{(1)}{=} \begin{vmatrix} a & b \\ c & d \end{vmatrix} - \ell \begin{vmatrix} c & d \\ c & d \end{vmatrix}$$

$$\overset{(4)}{=} \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

6) $A \xrightarrow{\text{Elim}} U$ in echelon form

$\Rightarrow \det A = \pm \det U$

number of row exchanges

7) If A has a row of zeros, then $\det A = 0$.

$$\begin{vmatrix} a & b \\ 0 & 0 \end{vmatrix} = a \cdot 0 - b \cdot 0 = 0.$$

<u>Reason</u>: Multiply the row of zeros by 2 $\Rightarrow$ $\det A = 2 \cdot \det A$ (3), (2)

$\Rightarrow \det A = 0$.

8) If A is triangular,

$\det A = a_{11} \cdot a_{22} \cdot \cdots \cdot a_{nn}$.

<u>Case 1</u>: No zeros on the diagonal.

$$A \xrightarrow{\text{Elim}} D = \begin{bmatrix} a_{11} & & 0 \\ & a_{22} & \\ 0 & & \ddots \; a_{nn} \end{bmatrix}$$

$\det A = \det D$ (5)

$= a_{11} a_{22} \cdots a_{nn} \cdot \det I$ (3)

$= a_{11} a_{22} \cdots a_{nn}$ (1).

<u>Case 2</u>: At least one zero on the diagonal. Elimination produces row of zeros $\Rightarrow \det A = 0$ by (6).

9) $\det(AB) = \det(A) \cdot \det(B)$.

<u>Reason</u>: The hardest case is when $\det B \neq 0$. Then we define

$$d(A) := \frac{\det(AB)}{\det(B)}$$

and check that $d(A)$ satisfies (1), (2) and (3).

10) $\det(A^{-1}) = 1 / \det(A)$.

11) $\det A^T = \det A$

<u>Reason</u>: $PA = LU \Rightarrow A^T P^T = U^T L^T$

$\det P = \pm 1$, $PP^T = I$